NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS | DT: 19/01/2026
Time: 3 Hrs | GTM – 19 | Max. Marks: 300

# KEY SHEET

## PHYSICS

| 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 |
|---|---|---|---|---|---|---|---|---|---|
| B | D | C | D | A | C | B | D | D | C |
| 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
| B | D | A | A | A | A | A | A | C | B |
| 21 | 22 | 23 | 24 | 25 | | | | | |
| 9 | 5 | 8 | 2 | 5 | | | | | |

## CHEMISTRY

| 26 | 27 | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 |
|---|---|---|---|---|---|---|---|---|---|
| D | D | B | C | B | A | A | C | B | D |
| 36 | 37 | 38 | 39 | 40 | 41 | 42 | 43 | 44 | 45 |
| B | A | B | A | C | D | B | D | A | C |
| 46 | 47 | 48 | 49 | 50 | | | | | |
| 5 | 4 | 121 | 16 | 32 | | | | | |

## MATHEMATICS

| 51 | 52 | 53 | 54 | 55 | 56 | 57 | 58 | 59 | 60 |
|---|---|---|---|---|---|---|---|---|---|
| B | A | C | B | D | A | B | C | B | D |
| 61 | 62 | 63 | 64 | 65 | 66 | 67 | 68 | 69 | 70 |
| B | D | B | C | A | A | D | C | A | D |
| 71 | 72 | 73 | 74 | 75 | | | | | |
| 2 | 1 | 21 | 12 | 90 | | | | | |

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# HINTS
# PHYSICS

1. Let final velocity is $V_x\hat{i}+V_y\hat{j}$

Then $V_x=V_0$, and $t=\dfrac{l}{V_0}$

$$V_y=a_yt=\frac{eEl}{mV_0};\ \text{where}\ \left(a_y=\frac{eE}{m}\right)$$

$$\therefore \tan\theta=\frac{V_y}{V_x}=\frac{eEl}{mV_0^2}=\frac{1.6\times10^{-19}\times91\times10^{-6}\times1}{9.1\times10^{-31}\times16\times10^{6}}=1\Rightarrow\theta=45^\circ$$

2. Conceptual

3. Moment of inertia of complete disc of mass $9m$ and radius $R$, about an axis passing through its centre and perpendicular to its plane can be written as follow:

$$I=\frac{1}{2}(9m)R^2=\frac{9}{2}mR^2$$

We can assume distribution of mass proportional to area. And area of the hole is $\dfrac{1}{9}$ times to that of area of complete disc. Hence mass of removed portion can be assume to be $m$. Moment of inertia of hole can be written about the same axis using parallel axis theorem. $I_{hole}=I_{cm}+md^2=\dfrac{1}{2}m\left(\dfrac{R}{3}\right)^2+m\left(\dfrac{2R}{3}\right)^2$

$$=\frac{1}{2}mR^2$$

Hence moment of inertia of the remaining system can be calculate as follows:

$$I_{remaining}=I-I_{hole}=\frac{9}{2}mR^2-\frac{1}{2}mR^2=4mR^2$$

4. The process goes as shown in this diagram

Ice 0°C —$Q_1$— Water 0°C —$Q_2$— Water 100°C —$Q_3$— Steam 100°C

$\Rightarrow Q_1=$ Latent heat required to convert $1gm$ ice to water at $0°C=$ Latent heat of fusion of water = 80 Cal



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$\Rightarrow Q_2 =$ Heat required to raise the temperature from $0°C$ to $100°C$ of $1gm$ water

$= m.s\Delta t = 1\times1\times(100°C - 0°C) = 100\ Cal$

$\Rightarrow Q_3 =$ Latent heat required to convert $1gm$ of water to steam $100°C$ of

1 gm=536 Cal

Total heat energy required

$Q = Q_1 + Q_2 + Q_3 = 80\ Cal + 100\,Cal + 536\,Cal = 716\,Cal$

So, the total heat required for the given process is $716\ Cal$

5. For the second orbit $(n=2)$

$2\pi r_2 = 2\lambda_2$

For the third orbit $(n=3)$

$2\pi r_3 = 3\lambda_3$

$\frac{\lambda_2}{\lambda_3} = \frac{\pi r_2}{\frac{2\pi r_3}{3}} = \frac{3r_2}{2r_3}$

The radius of the $n^{th}$ orbit in a hydrogen atom is given by $r_n \propto \frac{n^2}{Z}$

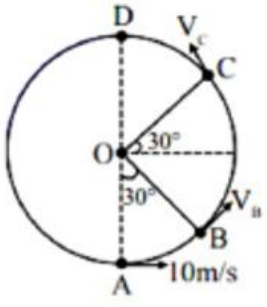
Where $Z$ is the atomic number. For hydrogen, $Z=1$, so,

$r_2 \propto 2^2 = 4$

$r_3 \propto 3^2 = 9$

Now we substitute $r_2$ and $r_3$ into the ratio $\frac{\lambda_2}{\lambda_3} = \frac{3.4}{2.9} = \frac{12}{18} = \frac{2}{3}$

6.






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$$\frac{1}{2}m\times 100+0=\frac{1}{2}mV_B^2+mg\left(R-\frac{R\sqrt{3}}{2}\right)$$

$$100=V_B^2+2gR\left(1-\frac{\sqrt{3}}{2}\right)$$

$$V_B^2=100-20\left(2-\sqrt{3}\right)$$

$$V_B^2=\left(60+20\sqrt{3}\right)$$

$$K.E_B=\frac{1}{2}mV_B^2=\frac{m}{2}\left(60+20\sqrt{3}\right) \quad \text{.....(1)}$$

$$\frac{1}{2}m(100)=\frac{1}{2}mV_c^2+mg\left(\frac{3R}{2}\right)$$

$$100=V_C^2+60$$

$$V_C^2=40$$

$$K.E_C=\frac{1}{2}mV_C^2=\frac{1}{2}m(40) \quad \text{.....(2)}$$

$$\frac{K.E_B}{K.E_C}=\frac{60+20\sqrt{3}}{40}$$

$$=\frac{3}{2}+\frac{\sqrt{3}}{2}=\frac{3+\sqrt{3}}{2}$$

7. The focal length $f$ of the equivalent mirror is

$$-\frac{1}{f}=\frac{2}{f_1}+\frac{1}{f_m}=\frac{2}{15}+\frac{1}{\infty}\Rightarrow f=-\frac{15}{2}cm$$

Since $f$ has a negative value, the combination behaves as a converging mirror.

Here $u=-20cm,\ f=-\frac{15}{2}cm,\ v=?$

According to mirror formula $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$



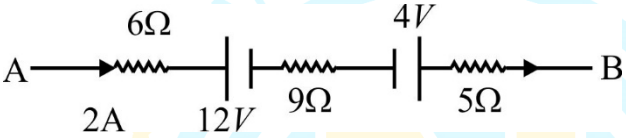
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$$\Rightarrow \frac{1}{v}-\frac{1}{-20}=\frac{1}{\frac{-15}{2}} \Rightarrow v=-12cm$$

Negative sign indicates that the image is $12\ cm$ in front of mirror

8. A diode is said to be forward biased if $p-$ side is at higher potential than $n-$ side of $p-n$ junction.

9. This is a series connection. Further whatever current enters A has to pass B. $I=2A$




The total resistance $=6+9+5=20\Omega$ The effective potential across the resistances is $20\Omega\times 2A=40V$. But $(+12-4)V$ is opposing the potential difference across AB therefore the potential difference applied across AB is $40V+8V=48V$

10. $F_1=\frac{GMm}{18^2}$

Density of big sphere $\rho=\frac{M}{\frac{4}{3}\pi R^3}$. So mass of cavity

$$m_c=M\left(\frac{r^3}{R^3}\right)=60.\frac{2^3}{6^3}=60.\frac{8}{216}=60.\frac{1}{27}=\frac{20}{9}$$

Force due to the (would-be) cavity mass at $14m$

$$F_c=\frac{Gm_cm}{14^2}$$

Because the cavity is removed, its attraction is subtracted from $F_1$. Hence

$$F_2=F_1-F_c$$

$$\frac{F_c}{F_1}=\frac{m_c}{M}.\frac{18^2}{14^2}=\frac{1}{27}.\frac{324}{196}=\frac{324}{5292}=\frac{3}{49}$$

$$F_2=F_1\left(1-\frac{3}{49}\right)=F_1.\frac{46}{49}$$

Therefore the ratio

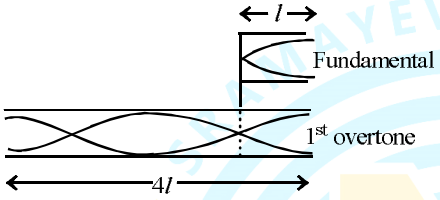


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$$F_1 : F_2 = 1 : \frac{46}{49} = 49 : 46$$

11. Closed –Open

$$\frac{V}{4l_1} = \frac{V}{l_2}$$




$$l_2 = 4l_1$$

$$l_1 = \frac{l_2}{4} = \frac{50}{4} = 12.5cm$$

12. We know that, $\varepsilon_0$ is vacuum permittivity.

The unit of vacuum permittivity is $kg^{-1}m^{-3}A^2s^4$

The dimension of vacuum permittivity is $\left[M^{-1}L^{-3}A^2T^4\right]$

Also, we know that, $\mu_0$ is permeability.

The unit of permeability is $kg.m.s^{-2}A^{-2}$

The dimension of permeability is $\left[MLA^{-2}T^{-2}\right]$

The dimensions of $\frac{\varepsilon_0}{\mu_0} = \frac{\left[M^{-1}L^{-3}A^2T^4\right]}{\left[MLA^{-2}T^{-2}\right]}$

$$\Rightarrow \frac{\varepsilon_0}{\mu_0} = \left[M^{-2}L^{-4}A^4T^6\right]$$

$$\therefore \sqrt{\frac{\varepsilon_0}{\mu_0}} = \left[M^{-1}L^{-2}A^2T^3\right]$$

13. $\phi = \frac{q_{enclosed}}{\varepsilon_0}$

14. 20 MSD = 1 cm



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$1\text{MSD} = \frac{1}{20} cm$

$10VSD = 9MSD$

$1VSD = \frac{9}{10} MSD$

$= \frac{9}{10} \times \frac{1}{20} cm$

$IVSD = \frac{9}{200} cm$

$VC = 1MSD - 1VSD$

$= \frac{1}{20} cm - \frac{9}{200} cm$

$= \frac{1}{200} \times 10mm$

$VC = 5 \times 10^{-2} mm$

15. Photon energy:

$$E = \frac{1240}{550} \approx 2.25eV$$

Compare with work functions:

$Cs : 1.9\ eV \rightarrow 2.25 > 1.9 \rightarrow$ Yes

$Li : 2.5\ eV \rightarrow 2.25 < 2.5 \rightarrow$ No

16. The energy radiated per sec is given by $E = e\sigma AT^4$ for same material $e$ is same

$$\frac{E_1}{E_2} = \frac{T_1^4 A_1}{T_2^4 A_2} = \frac{T_1^4 \left(4\pi r_1^2\right)}{T_2^4 \left(4\pi r_2^2\right)} = \frac{(4000)^4 \times 1^2}{(2000)^4 \times 4^2} = \frac{1}{1}$$

17. Since the refractive index of the liquid is greater than the refractive index of glass, therefore the focal length of the system has to be negative. So, Options (b) and (d) are excluded.

Now, $\frac{1}{24} = (1.5-1)\frac{2}{R}$ (or) $R = 24cm$

Again, for liquid concave lens, $\frac{1}{f} = -(1.6-1)\left(\frac{2}{24}\right)$

$\frac{1}{f} = -\frac{3}{5} \times \frac{1}{12}$ (or) $\frac{1}{f} = -\frac{1}{12}$ (or) $f = -20cm$



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Now, $\frac{1}{F}=\frac{1}{24}-\frac{1}{20}=\frac{5-6}{120}$ or $F=-120cm$

18. $\beta=\frac{\lambda D}{d}$

In water, $\lambda$ decreases $\rightarrow \beta$ decreases.

Both statements true, and $R$ correctly explains $A$.

19. As the potential difference is constant hence we can say that

$Q_1=60\mu C=V\times C$ ….(1)

Now there is already $60\mu C$ on the capacitor.

More $120\mu C$ charge flows from battery. Hence net charge on capacitor is

$Q_2=180\mu C=V\times KC$ ….(2)

(2)/(1) $\Rightarrow$ 3= $K$

20. Resistivity decreases with temperature in semiconductors

21. Let object distance $u=90-10t\ (cm)$ and detector distance $d=10t\ (cm)$.

Focal length $f=20cm$

Image distance $v=\frac{uf}{u-f}$. For maximum intensity $v=d$

$$\frac{(90-10t).20}{(90-10t)-20}=10t$$

Simplify: $10t^2-90t+180=0\Rightarrow t^2-9t+18=0$

$t=3,6$

22. For bubbles in contact (curvature toward smaller bubble):

$$\frac{1}{R}=\frac{1}{r_1}-\frac{1}{r_2}=\frac{1}{4}-\frac{1}{20}=\frac{1}{5}\Rightarrow R=5\ cm$$

23. $AM=m\vec{r}\times\vec{V}=2\left(\hat{i}+\hat{j}\right)\times(2)\left(\hat{i}-\hat{j}+\hat{k}\right)$

$=-8\hat{k}-4\hat{j}+4\hat{i}$

$\therefore$ about $z$ axis= $-8$



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24. $R_{eq} = R_1 + R_2 = \frac{x}{kA} + \frac{4x}{2kA} = \frac{3x}{kA}$

$$\frac{dQ}{dt} = \frac{\Delta T}{\frac{3x}{kA}} = \frac{(T_2 - T_1)kA}{3x} = \frac{1}{3}\left[\frac{A(T_2 - T_1)k}{x}\right]$$

25. $u = 0$

$$s = \frac{1}{2}at^2$$

$$= \frac{1}{2} \times 10 \times 9 = 45m$$

$$S_n = u + a\frac{(2n-1)}{2}$$

$$45 = \frac{10}{2}(2t - 1)$$

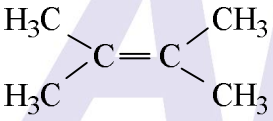
$$10 = 2t$$

$$t = 5 \text{ sec}$$

## CHEMISTRY

26. $r = \frac{53 \times n^2}{Z} = 27\ pm$

27. A, B, C exhibit geometrical isomerism.



28. Changing water from liquid to vapour.

29. Fluorine is the most electronegative element.

30. The size of trivalent ions decreases from left to right, this is called lanthanide contraction.

31. 4g of propyne gives 0.05 mole of $H_2$

32. Benzaldehyde can't give Fehilings test

33. $CH_3COONa$ on electrolysis gives two gases $C_2H_6$ & $CO_2$ at anode.



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34. Allyl carbocation is stable. It gives $S_N1$

35. Only thiamine is a vitamin

36. Electronegativity ; B > Tl > In > Ga > Al

Melting point ; C > Si > Ge > Sn < Pb

Ionic radius ; C < Si < Ge < Sn < Pb

Ionisation potential ; B > Ga > Tl > Al > In

37. $\frac{A}{n} = A.e^{-\lambda t} \Rightarrow \frac{1}{n} = e^{-\lambda t}$

$$\ln\frac{1}{n} = -\lambda t \Rightarrow -\ln n = -\lambda t;\ \lambda = \frac{\ln n}{t}$$

38. Major product is biphenyl

39. Five isomers are possible (2 inactive + 3 active)

40. $A + 3B \rightleftharpoons 4C$

$1-x\ \ 1-3x\ \ 4x$

At eqn $1-x = 4x$

$x = 0.2$

41. Eq.volume of $H_2$ + Eq.volume of $O_2$ =1.68L

During electrolysis $H_2$ and $O_2$ are liberated in 2:1 volume ratio.

$\therefore$volume of $H_2$ = 1.12L

Equivalents of $H_2$ = 0.1= eq. of $MnO_4^-$

$\therefore$Moles of $MnO_4^-$ = 0.02

42. N-Ethyl-2-methyl butanamide

43. Conceptual

44. The value of van't Hoff's factor for the solutions increases with dilution.

45. $[Ni(CO)_4]$ is $sp^3$ hybridised and tetrahedral

$[Ni(CN)_4]^{2-}$ is $dsp^2$ hybridised and square planar

$[NiCl_4]^{2-}$ is $sp^3$ hybridised and tetrahedral

46. $XeO_2F_2$ $XeOF_2$, $XeF_2$, $ClF_3$, $PCl_4^-$ have $sp^3d$ hybridisation.

47. Let $K_1$, $Ea_1$ & $T_1$ are rate constant, activation energy and temperature of uncatalysed reaction. $K_2$, $Ea_2$ & $T_2$ are rate constant, activation energy and temperature of catalysed reaction, then $K_1 = Ae^{-Ea_1/RT_1}$ ; $K_2 = Ae^{-Ea_2/RT_2}$

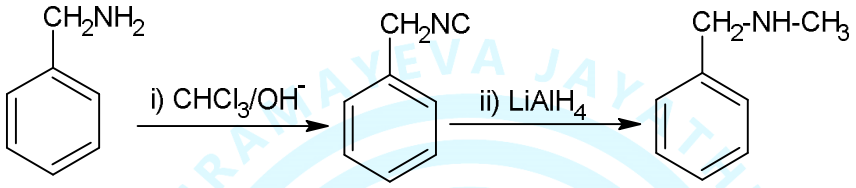


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Given, $K_1 = K_2$ *i.e.* $Ae^{-Ea_1/RT_1} = Ae^{-Ea_2/RT_2}$

$$\frac{Ea_1}{RT_1} = \frac{Ea_2}{RT_2}$$

48.




49. Let the normality of $H_2O_2$ solution be $N_x$

No. of milliequivalents of $Na_2S_2O_3$ = No. of milliequivalents of $H_2O_2$

40 x 0.1 = 25 x $N_x$

$$N_x = \frac{40 \times 0.1}{25} = 0.16N$$

50. weight of Substance. 0.2gm

weight of barium Sulphate = 0.466gm

weight percentage of Sulphur = $\frac{0.466 \times 32 \times 100}{0.2 \times 233}$

## MATHEMATICS

51. $a_3.a_5 = \left(a.r^2\right)\left(a.r^4\right) = a^2.r^6 = 729 = 27^2 \Rightarrow a_4 = a.r^3 = 27$

$$a_2 = \frac{111}{4} - 27 = \frac{3}{4}$$

$$r^2 = \frac{4.27}{3} = 4 \times 9 = 36 \Rightarrow r = 6 \Rightarrow a = \frac{3}{24} = \frac{1}{8}$$

$$24\left(a_1 + a_2 + a_3\right) = 24\left(a + ar + ar^2\right)$$

$$= 24 \times \frac{1}{8}\left(1 + 6 + 6^2\right) = 3 \times \left(1 + 6 + 36\right) = 3 \times 43 = 129$$

52. $\Rightarrow \frac{dy}{dx} + 3\left(\sec^2 x\right)y = \sec^2 x$

I.F $= e^{\int 3\sec^2 x dx} = e^{3\tan x}$



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$$y.e^{\tan x} = \int e^{3\tan x}.\sec^2 x dx + c$$

$$y.e^{3\tan x} = \frac{e^{3\tan x}}{3} + c$$

Also $f(0) = \frac{1}{3} + e^3 \Rightarrow \left(\frac{1}{3} + e^3\right) = \frac{1}{3} + c \Rightarrow c = e^3$

$$\therefore y.e^{3\tan x} = \frac{e^{3\tan x}}{3} + e^3$$

Put $x = \frac{\pi}{4}$: $ye^3 = \frac{e^3}{3} + e^3 \Rightarrow y = \frac{4}{3}$

53. $P(1R \text{ and } 1B) = P(A).P\left(\frac{1R1B}{A}\right) + P(B).P\left(\frac{1R1B}{B}\right)$

$$= \frac{1}{2}.\frac{{}^3C_1.{}^1C_1}{{}^6C_2} + \frac{1}{2}.\frac{{}^2C_1.{}^3C_1}{{}^{n+5}C_2}$$

$$P\left(\frac{1R1B}{A}\right) = \frac{\frac{1}{2}.\frac{3}{15}}{\frac{1}{2}.\frac{3}{15} + \frac{1}{2}.\frac{6.2}{(n+5)(n+4)}} = \frac{6}{11}$$

$\Rightarrow n^2 + 9n - 52 = 0 \Rightarrow n = 4$

54. $\left(\log_{2026} x\right)^2 = 2\log_{2026} x - \frac{1}{2} \Rightarrow 2t^2 - 4t + 1 = 0$ ; where $t = \log_{2026} x$

$\Rightarrow \log_{2026} x_1 + \log_{2026} x_2 = 2 \Rightarrow x_1 x_2 = (2026)^2$

55. Area $\Delta ACD = \frac{1}{2}\begin{vmatrix} \frac{3}{\sqrt{a}} & \sqrt{a} \\ -\frac{3}{\sqrt{a}} & -\sqrt{a} \\ 3\cos\theta & a\sin\theta \end{vmatrix}$

$$= 3\sqrt{a}\left|\cos\theta - \sin\theta\right|$$



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Max value of function is $3\sqrt{a}\sqrt{2}=12\Rightarrow 2a=16\Rightarrow a=8$

56. $11\alpha(10,6)+18\alpha(11,5)=p(14)^6$

$$=11\int_0^2 t^{10}(1+3t)^6\,dt+18\int_0^2 t^{11}(1+3t)^5\,dt$$

$$=11\left[(1+3t)^6.\frac{t^{11}}{11}-\int 6(1+3t)^5.3\frac{t^{11}}{11}\right]_0^2+18\int_0^2 t^{11}(1+3t)^5\,dt$$

$$=\left(t^{11}(1+3t)^6\right)_0^2=32(14)^6$$

57. Solve both equations

$y=\frac{x^2}{k}-5$ and $x^2+y^2=25$

We get $A(0,-5), B\left(-\sqrt{10k-k^2},5-k\right), C\left(\sqrt{10k-k^2},5-k\right)$

Area of $\Delta=\left(\sqrt{10k-k^2}\right)|10-k|$

$10k-k^2=p^2\,(p\in I)\Rightarrow (k-5)^2+p^2=25$

$k-5=0$

$k-5=\pm 3$

$k-5=\pm 4$

$k=1,2,5,8,9$

58. $\vec{b}_1\times\vec{b}_2=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\1&-a&0\\1&-1&1\end{vmatrix}=-a\hat{i}-\hat{j}+(a-1)\hat{k}$

$\vec{a}_1-\vec{a}_2=-\hat{i}+\hat{j}+\hat{k}$

Shortest distance $=\left|\frac{(\vec{a}_1-\vec{a}_2).(\vec{b}_1\times\vec{b}_2)}{|\vec{b}_1\times\vec{b}_2|}\right|$



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$$\Rightarrow \sqrt{\frac{2}{3}} = \left| \frac{2(a-1)}{\sqrt{a^2+1+(a-1)^2}} \right|$$

$$\Rightarrow 6\left(a^2-2a+1\right) = 2a^2-2a+2$$

$$\Rightarrow (a-2)(2a-1) = 0 \Rightarrow a = 2 \text{ because } a \in Z$$

59. Given $f(x+y) = f(x).f(y) \forall x, y \in R$

$\therefore f(x) = a^x$

$f'(x) = a^x.\log(a)$

Now, $f'(0) = \log(a) \Rightarrow 3 = \log(a) \Rightarrow a = e^3$

$\therefore f(x) = e^{3x}$

Now, $\lim\limits_{h\to 0}\left(\frac{f(h)-1}{h}\right) = \lim\limits_{h\to 0}\left(\frac{e^{3h}-1}{h}\right) = \lim\limits_{h\to 0}\left(\frac{e^{3h}-1}{3h}\times 3\right) = 3$

60. Case $-1$: All digits are non-zero

${}^9C_2.2.\frac{3!}{2} = 216$

Case $-2$: One digit is 0

$0,0,x \Rightarrow {}^9C_1.1 = 9$

$0,x,x \Rightarrow {}^9C_1.2 = 18$

Total $= 216+27 = 243$

61. Given, $\left(\sin^{-1}x\right)^2 - \left(\cos^{-1}x\right)^2 = a$

$\Rightarrow \left(\sin^{-1}x+\cos^{-1}x\right)\left(\sin^{-1}x-\cos^{-1}x\right) = a$

$\Rightarrow \frac{\pi}{2}\left(\sin^{-1}x-\cos^{-1}x\right) = a \Rightarrow \frac{\pi}{2} - 2\cos^{-1}x = \frac{2a}{\pi} \Rightarrow 2\cos^{-1}x = \frac{\pi}{2} - \frac{2a}{\pi}$



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$$\Rightarrow \cos^{-1}\left(2x^2-1\right)=\frac{\pi}{2}-\frac{2a}{\pi} \Rightarrow 2x^2-1=\cos\left(\frac{\pi}{2}-\frac{2a}{\pi}\right)$$

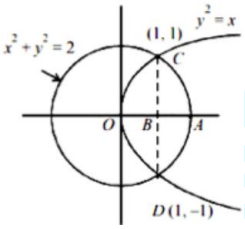
$$\Rightarrow 2x^2-1=\sin\left(\frac{2a}{\pi}\right)$$

62. $$R_2=\int_{-2}^{3} xf\left(x\right)dx=\int_{-2}^{3}\left(1-x\right)f\left(1-x\right)dx \Rightarrow R_2=\int_{-2}^{3}\left(1-x\right)f\left(x\right)dx$$

$$\therefore R_2+R_2=\int_{-2}^{3} df\left(x\right)dx+\int_{-2}^{3}\left(1-x\right)f\left(x\right)dx=\int_{-2}^{3} f\left(x\right)dx=R_1$$

$$\Rightarrow R_1=2R_2$$

63.




Area of $OCADO=2\int_{0}^{1}\sqrt{x}dx+2\int_{1}^{2}\sqrt{2-x^2}dx$

$$=2\left[\frac{2}{3}.1-0\right]+2\left[\frac{x\sqrt{2-x^2}}{2}+\sin^{-1}\frac{x}{\sqrt{2}}\right]_1^{\sqrt{2}}=\frac{3\pi+2}{6}$$

Remaining area $=2\pi-\frac{3\pi+2}{6}=\frac{9\pi-2}{6}$

$\therefore$ Required ratio $=\frac{9\pi-2}{3\pi+2}$

64. Let $P=\left(0,3+\sqrt{5}\right)$ and foci $S=\left(2,3\right),S'=\left(-2,3\right)$

$$PS=\sqrt{\left(-2\right)^2+\left(\sqrt{5}^2\right)}=\sqrt{9}=3, PS'=\sqrt{2^2+\left(\sqrt{5}\right)^2}=3 \Rightarrow PS+PS'=6=2a$$

$$\Rightarrow a=3$$



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65. $$T_n = \frac{(2n+5)}{(n+1)(n+2)} \times \left(\frac{1}{3}\right)^n$$

$$= \frac{1}{n+1} \times \frac{1}{3^{n-1}} - \frac{1}{n+2} \times \frac{1}{3^n}$$

$$\sum_{n=1}^{10} T_n = \frac{1}{2} - \frac{1}{12 \times 3^{10}}$$

66. $\underset{0}{HHH}$ $H\underset{1}{HT}$ $\underset{1}{HT}H$ $\underset{0}{THH}$ $\underset{1}{HTT}$ $T\underset{1}{HT}$ $\underset{0}{THH}$ $\underset{0}{TTT}$

$\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$ $\frac{1}{8}$

| $X = x$ | 0 | 1 |
|---|---|---|
| $P(X = x)$ | 4/8 | 4//8 |

$$\mu = \sum_i x_i P(X = x_i) = \frac{4}{8} = \frac{1}{2}$$

$$\sigma^2 = \sum_i (x_i P(X = x_i)) - (\mu^2) = 0 + \frac{1}{8} + 0 + \frac{1}{8} + \frac{1}{8} - \frac{1}{4} = \frac{1}{4}$$

$$\therefore 64(\mu + \sigma^2) = 64\left(\frac{1}{2} + \frac{1}{4}\right) = 48$$

67. $3,7,9 \rightarrow$ odd and $0,4,6,8,10 \rightarrow$ even

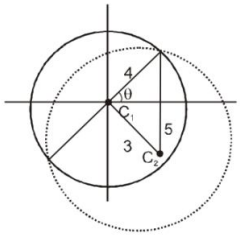
$${}^3C_1 \cdot {}^5C_1 = 15 + (6,4),(8,6),(10,8),(9,7)$$

Minimum ordered pairs to be added = $15 + 4 = 19$



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68.




Slope of $C_1C_2$ is $\tan\alpha = -\frac{4}{3}$

By using parametric coordinates

$C_2\left(\pm 3\cos\alpha, \pm 3\sin\alpha\right)$

$C_2\left(\pm 3(-3/5), \pm 3(4/5)\right)$

$C_2\left(\pm 9/5, \mp 12/5\right)$

69. Sum of all elements of $A \cap B = 2$ [sum of natural numbers upto 100 which are neither divisible by 3 nor by 5]

$$= 2\left[\frac{100\times 101}{2} - 3\left(\frac{33\times 34}{2}\right) - 5\left(\frac{20\times 21}{2}\right) + 15\left(\frac{6\times 7}{2}\right)\right]$$

$$= 10100 - 3366 - 2100 + 630 = 5264$$

70. $z_1 = \sqrt{3} + 2\sqrt{2}i$ & $\frac{|z_2|}{|z_1|} = \frac{1}{\sqrt{3}}$, $z_2 = \frac{|z_2|}{|z_1|}.z_1 e^{i\left(\frac{\pi}{6}\right)}$

$$z_2 = \frac{\left(3-2\sqrt{2}\right) + i\left(2\sqrt{6}+\sqrt{3}\right)}{2\sqrt{3}}$$

$$z_1 - z_2 = \frac{\left(3+2\sqrt{2}\right) + i\left(2\sqrt{6}-\sqrt{3}\right)}{2\sqrt{3}}$$

$|z_1 - z_2| = |z_2| \Rightarrow \Delta ABO$ is isosceles with angles $\frac{\pi}{6}, \frac{\pi}{6}$ & $\frac{2\pi}{3}$

71. $M^{-1}adj\left(adj\ M\right) = k^2 I$



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$$\left|adj\left(adj\ M\right)\right| = \left|k^2 M\right|$$

$$\left|M\right|^{(3-1)^2} = k^6 \left|M\right| \Rightarrow k^2 = 4$$

72. Using result

$$\sum_{r=0}^{n} r^2 .{}^nC_r = n(n+1).2^{n-2}$$

Then $\sum_{r=0}^{2026} r^2 .{}^{2026}C_r = 2026 \times 2027 \times 2^{2024}$

$= 2026 \times \alpha \times 2^{2024} \Rightarrow \alpha = 2027$ = Prime Number

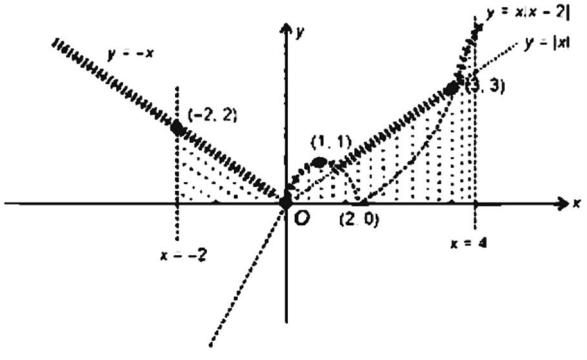
73. $\left|\vec{a} \times \vec{b}\right|^2 + \left|\vec{a}.\vec{b}\right|^2 = \left|\vec{a}\right|^2 .\left|\vec{b}\right|^2$

$$\Rightarrow 5 + 9 = 6\left|\vec{b}\right|^2 \Rightarrow \left|\vec{b}\right|^2 = \frac{7}{3}$$

$$\left|\vec{a} - \vec{b}\right| = \sqrt{\left|\vec{a}\right|^2 + \left|\vec{b}\right|^2 - 2\vec{a}.\vec{b}} = \sqrt{\frac{7}{3}}$$

Projection of $\vec{b}$ on $\vec{a} - \vec{b} = \frac{\vec{b}.\left(\vec{a} - \vec{b}\right)}{\left|\vec{a} - \vec{b}\right|} = \frac{\vec{b}.\vec{a} - \left|\vec{b}\right|^2}{\vec{a} - \vec{b}} = \frac{3 - \frac{7}{3}}{\sqrt{\frac{7}{3}}} = \frac{2}{\sqrt{21}}$

74.




Area = $\frac{1}{2} \times 2 \times 2 + \int_0^1 \left(2x - x^2\right)dx + \int_1^3 x\,dx + \int_3^4 \left(x^2 - 2x\right)dx$

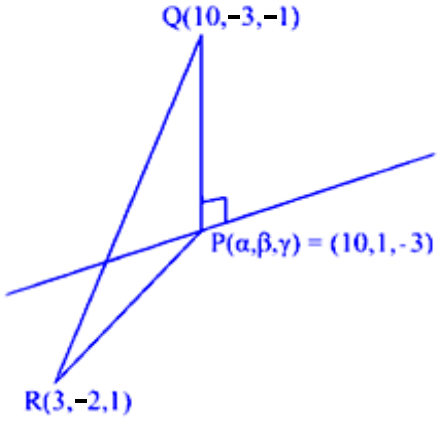


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$$= 2 + \left[\frac{2x^2}{2} - \frac{x^3}{3}\right]_0^1 + \left[\frac{x^2}{2}\right]_1^3 + \left[\frac{x^3}{3} - \frac{2x^2}{2}\right]_3^4$$

$= 12$ square units

75.




$$\frac{x-3}{7} = \frac{y-2}{-1} = \frac{z+1}{-2} = \lambda$$

$$\Rightarrow (7\lambda + 3, -\lambda + 2, -2\lambda - 1)$$

Dr's of $QP \Rightarrow (7\lambda - 7, -\lambda + 5, -2\lambda)$

Now, $(7\lambda - 7).7 - (-\lambda + 5) + (2\lambda).2 = 0 \Rightarrow 54\lambda - 54 = 0 \Rightarrow \lambda = 1$

$\therefore P = (10,1,-3) \Rightarrow \overrightarrow{PQ} = -4\hat{j} + 2\hat{k}$, $\overrightarrow{PR} = -7\hat{i} - 3\hat{j} + 4\hat{k}$

$$\text{Area} = \left|\frac{1}{2}\begin{vmatrix} i & j & k \\ 0 & -4 & 2 \\ -7 & -3 & 4 \end{vmatrix}\right| = 3\sqrt{30}$$



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